

# From qualitative prediction to quantitative insight: combined meteorological patterns and regional dynamics of severe fever with thrombocytopenia syndrome in Liaoning Province, China, 2010–2024

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## ABSTRACT

**Background:** Severe fever with thrombocytopenia syndrome (SFTS) is an emerging tick-borne disease with an expanding range and increasing public health burden. Meteorology-driven frameworks that integrate qualitative prediction with quantitative risk estimation while accommodating lag, regional heterogeneity, autoregressive case count effects, and zero-inflated counts remain scarce.

**Methods:** Monthly SFTS case counts and meteorological data from thirteen prefecture-level cities in Liaoning Province, China, from 2010 to 2024 were analyzed. Fushun was excluded because all counts were zero. Predictors were screened by correlation and variance inflation factor (VIF), and Boruta plus conditional permutation importance selected nine variables. Cities were grouped by k-means clustering. Four algorithms, including random forest (RF), extreme gradient boosting (XGBoost), gradient boosting decision tree (GBDT), and light gradient boosting machine (LightGBM), classified case presence using 2010–2022 training with ten-fold cross-validation and 2023–2024 testing. Shapley additive explanations (SHAP) interpreted variable importance and lagged associations in Dalian and Dandong. A mixed generalized additive model (MGAM) with distributed lag nonlinear modeling (DLNM) estimated exposure-lag effects of each meteorological main exposure.

**Results:** Nine meteorological variables were retained: wind speed (WS), relative humidity (RH), precipitation (PRCP), air pressure (AP), sunshine duration (SD), diurnal temperature range (DTR), surface air temperature difference (STD), standardized precipitation evapotranspiration at one month (SPEI1), and six months (SPEI6). K-means clustering grouped the thirteen Liaoning cities into three climatic groups. Across four classifiers, RF performed best in high-incidence areas, XGBoost was most stable; SHAP revealed opposite lag effects for some variables, indicating nonlinear delayed influences. Quantitative risk estimation selected the optimal covariates for each main exposure, characterized exposure response shapes: inverted U for WS, AP, PRCP, DTR, and SPEI6; monotonic increase for RH and SD; monotonic decrease for STD; bimodal for SPEI1.

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**Conclusions:** This study identifies meteorological heterogeneity in high-incidence regions while quantifying province-wide risk windows for each meteorological exposure, thereby informing regional and provincial prevention and early warning strategies.

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## 1. Introduction

Severe fever with thrombocytopenia syndrome (SFTS) is an emerging tick-borne viral hemorrhagic fever caused by the severe fever with thrombocytopenia syndrome virus (SFTSV), also referred to as Bandavirus dabieense (Li et al., 2022; Yu et al., 2011). Since its identification in China in 2009, seasonally clustered cases have been reported across East Asia, including China, Japan, and South Korea (Kim et al., 2013; Li et al., 2018; Takahashi et al., 2014). Clinically, SFTS features fever with leukopenia and thrombocytopenia and may involve gastrointestinal and neurologic complications; the case-fatality risk varies markedly across regions and time. Ixodid ticks, particularly *Haemaphysalis longicornis*, are regarded as the principal vectors (Zhang et al., 2012), and healthcare-associated human-to-human transmission has been documented (Liu et al., 2012). The World Health Organization recognized SFTS as a priority emerging zoonosis requiring strengthened surveillance and research, given its strong climatic and ecological determinants and its continuing public health impact (Hu et al., 2018).

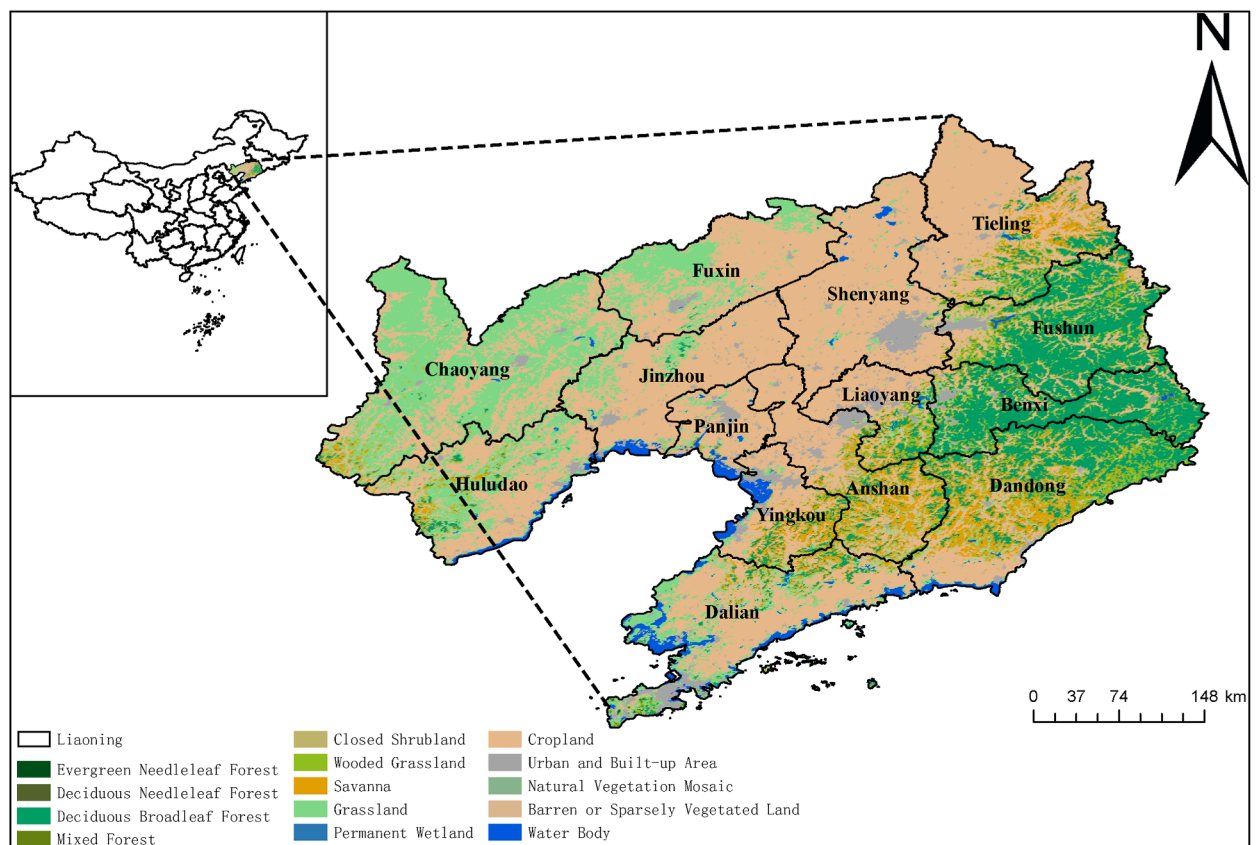
SFTS was first discovered in China in 2009 and was first reported in 2011 (Yu et al., 2011). The vast majority of SFTS cases were identified in Liaoning, Shandong, Anhui, Hubei, Henan, Zhejiang and Jiangsu Provinces. By 2018, cases had been documented across 25 provincial-level regions (Miao et al., 2021). In Liaoning Province, confirmed cases were initially concentrated in the eastern city of Dandong and the southern city of Dalian, whereas in recent years cases have spread to nearly all cities across the province. Accumulating evidence indicated that meteorological conditions, comprising temperature, relative humidity, precipitation, air pressure, wind speed, sunshine duration, drought, and wetness metrics such as the standardized precipitation evapotranspiration index (SPEI), along with diurnal temperature range, shape the seasonal timing and spatial distribution of SFTS (Ge et al., 2025; Wu et al., 2016, 2020). The environmental context further modulated these patterns, reflected in the normalized difference vegetation index (NDVI), land use, elevation (Li et al., 2012; Sun et al., 2018; Wang, Tian, et al., 2024; Zhang et al., 2019). Most SFTS studies have focused on quantitative risk estimation using time series generalized linear models (GLMs) (Wang et al., 2022), generalized additive models (GAMs) (Deng et al., 2022), and distributed lag nonlinear models (DLNMs) (Gasparrini, 2011), which characterize nonlinear and lagged associations. In contrast, machine-learning work has mainly targeted prediction and feature ranking, but is rarely linked to an inferential model that yields adjusted exposure–lag risk estimates. Few studies also derive exposure-specific parsimonious adjustment sets or explicitly model autoregressive dependence in case counts. We therefore propose a two-stage framework that combines machine-learning classification for rapid case-month signals and driver prioritization with MGAM–DLNM risk modeling incorporating autoregression and exposure-specific adjustment, improving both interpretability and practical utility compared with using either approach alone.

In this study, we aimed to develop a two-stage modeling framework for SFTS that links qualitative prediction with quantitative risk estimation. In the first stage, we applied machine learning classification to predict case presence, identified key meteorological drivers of high incidence in Dalian and Dandong, characterized heterogeneity between these areas, and depicted component-specific nonlinear and lag effects. These outputs provided fast signals for early warning and supported spatial prioritization of control resources. In the second stage, guided by the findings from the first stage, we fitted mixed generalized additive models (MGAMs) with distributed-lag terms and an autoregressive component for the cases. For each meteorological main exposure, we then selected a parsimonious, exposure-specific covariate set to isolate independent exposure–lag associations and to estimate clear risk windows and lag timing, improving interpretability and cross-exposure comparability. Together, the two stages delivered actionable evidence to time interventions and strengthen location-specific surveillance and preparedness.

## 2. Materials and methods

### 2.1. Study area

Liaoning Province is situated in the southern part of Northeast China, spanning latitude 38°43'N to 43°26'N and longitude 118°53'E to 125°46'E, and covers a total area of 148,600 km<sup>2</sup>. Administratively, the province consists of fourteen cities at the prefecture-level: Shenyang, Dalian, Anshan, Fushun, Benxi, Dandong, Jinzhou, Yingkou, Fuxin, Liaoyang, Panjin, Tieling, Chaoyang, and Huludao (Fig. 1). Liaoning is characterized by a temperate monsoon climate, with an average annual temperature of 7–11 °C and mean annual precipitation from 600 to 1100 mm. In this study, thirteen prefecture-level cities were included in the analysis, and Fushun was excluded due to the lack of SFTS case data.



**Fig. 1.** Map of Liaoning Province showing prefecture-level city boundaries and land-cover types. The inset map showed the location of Liaoning Province within China, while the main map depicted its fourteen prefecture-level cities. The legend illustrated the land-cover types across these cities. The base layer of the map was from the Resource and Environment Science and Data Center(<https://www.resdc.cn/DOI/DOI.aspx?DOIID=120>).

## 2.2. Data collection

Monthly SFTS case data from January 2010 to December 2024 were obtained from the Second Department of Infectious Diseases, First Affiliated Hospital of China Medical University. The dataset covered thirteen prefecture-level cities in Liaoning Province, excluding Fushun due to the absence of reported cases. The case records included the age, sex, residential address, month of disease onset, and clinical outcome (recovery or death) of the patients.

Meteorological data were collected for the same period and at the same spatial resolution as the SFTS case data. Monthly average atmospheric pressure (hPa), mean temperature (°C), maximum and minimum temperature (°C), land surface temperature (°C), cumulative precipitation (mm), average wind speed (m/s), mean gust speed (m/s), relative humidity (%), sunshine duration (h), and potential evapotranspiration (mm) were compiled. Land surface temperature data were obtained from the NASA GES DISC GLDAS\_NOAH025\_M\_2.1 product (<https://ldas.gsfc.nasa.gov/gldas/>), which provided global hourly values in Kelvin (K); these values were converted to degrees Celsius (°C) and rounded to two decimal places. All other meteorological variables were sourced from the ERA5 reanalysis dataset (<https://cds.climate.copernicus.eu/>) provided by the European Centre for Medium-Range Weather Forecasts (ECMWF). Monthly maximum and minimum temperatures were obtained by averaging daily maxima and minima, respectively.

Derived indices included the diurnal temperature range (DTR), calculated as the difference between the monthly average maximum and minimum temperatures, which reflected the amplitude of diurnal temperature variation, and the surface–air temperature difference (STD), reflecting urban heat island intensity. The standardized precipitation evapotranspiration index (SPEI) at multiple time scales (SPEI1, SPEI3, SPEI6, and SPEI12) was calculated using the “spei” package in R, based on cumulative precipitation and potential evapotranspiration, to quantify drought and wetness conditions.

Environmental data included monthly normalized difference vegetation index (NDVI), retrieved from NASA Earth Observations (<https://neo.gsfc.nasa.gov/>). Annual population data for each city were obtained from the Liaoning Statistical Yearbook (<https://tjj.ln.gov.cn/>).

### 2.3. Meteorological screening and clustering

Descriptive analyses were conducted to examine the temporal trends of SFTS cases and meteorological factors across the thirteen prefecture-level cities in Liaoning Province during the study period. Meteorological variables were initially screened using Spearman correlation analysis in conjunction with the variance inflation factor (VIF) assessment. For pairs of variables with correlation coefficients greater than 0.7 (Li et al., 2021), variables with VIF values exceeding 5 (O'Brien, 2007) were sequentially removed in descending order until multicollinearity was minimized. The remaining variables were further refined using Boruta feature selection (Kursa & Rudnicki, 2010) and conditional permutation importance (Keller, 2020) based on random forest, and were evaluated by ten-fold cross-validation to exclude uninformative meteorological factors. Based on the final set of selected variables, k-means clustering (Davis et al., 1998) was applied to classify the thirteen prefecture-level cities into three distinct groups according to their meteorological profiles.

### 2.4. Qualitative classification prediction

Binary classification analysis was conducted for two high-incidence cities, Dalian and Dandong, to predict the occurrence of SFTS cases based on meteorological exposures. To capture the lagged effects of meteorological variables on SFTS occurrence, a maximum lag of six months (Zhang et al., 2019) was incorporated, reflecting the ecological characteristics of tick vectors. Predictor variables included both contemporaneous and lagged meteorological factors, with SFTS case occurrence defined as a city-month binary outcome taking the value 1 when at least one case was reported in that month and 0 otherwise. Based on the k-means clustering of all cities, Dalian and Dandong were assigned to different meteorological groups; therefore, the cities were included as categorical covariates to account for this group-level heterogeneity in the pooled classification model. The qualitative analysis was restricted to Dalian and Dandong since they were the two high-incidence cities with sufficient case-month information to support reliable classification modeling. Four machine learning algorithms: random forest (RF), extreme gradient boosting (XGBoost), gradient boosting decision tree (GBDT), and light gradient boosting machine (LightGBM) were applied for classification. The datasets were partitioned into a training set (2010–2022) and a testing set (2023–2024). Model hyperparameters for RF, XGBoost, and GBDT were optimized via 10-fold cross-validation with a grid search, while LightGBM employed its own cross-validation method. Model performance was evaluated on the independent testing set using four primary metrics, including F1 score, AUC, precision, and recall.

Following model selection, interpretability was conducted using Shapley additive explanations (SHAP) (Wang, Liang, et al., 2024). SHAP values from the best-performing models were used to quantify the direction and magnitude of contribution for each predictor, with effects summarized by cluster and across lagged terms. Global and observation-level explanations were examined, and results were summarized to characterize heterogeneity in meteorological importance and lagged effects across two cities.

### 2.5. Quantitative risk estimation

In this study, a mixed generalized additive model (MGAM) framework was introduced to evaluate the associations between meteorological exposures and SFTS risk quantitatively. As an extension of the conventional generalized additive model, MGAM incorporates additive parametric functions for covariates, along with an autoregressive term, to account for short-term serial dependence (Ma et al., 2013). This specification effectively mitigates residual autocorrelation, yielding stable coefficient and standard error estimates. Cities were first grouped into three meteorological clusters by k-means, and the cluster membership was included in the MGAM as a random intercept to account for between-cluster heterogeneity while avoiding overparameterization from city-specific random effects. For model selection, each meteorological variable was sequentially designated as the main exposure and entered via a distributed-lag cross-basis. The remaining variables served as potential adjusters, also specified with distributed lag terms, to control concurrent meteorological conditions and potential confounding. For each main exposure, every non-empty subset of the other eight variables was considered, yielding 255 candidate adjustment sets and, consequently, 255 candidate models. These models were compared using the following steps:

- (1) Computing AIC for all candidate models and retaining those with  $\Delta\text{AIC} \leq 2$  (Burnham & Anderson, 2004) yielded a subset with essentially equivalent empirical support, where  $\Delta\text{AIC} = \text{AIC}_i - \min\{\text{AIC}_i\} (i = 1, 2, \dots, 255)$ .
- (2) Within this subset, selecting the model with the smallest BIC while also considering the significance of the risk estimates, thereby determining the most parsimonious specification.

The model was specified as follows:

$$\ln(\mathbb{E}[Y_i]) = \eta_i + b_{g(i)} + \omega_i, \quad (2.1)$$

where  $Y_i$  denotes the monthly SFTS case count for observation  $i$ ,  $\eta_i$  is the fixed-effects predictor,  $b_{g(i)}$  is the random intercept for the meteorological cluster  $g(i)$ , and  $\omega_i$  is the within-city autoregressive component. To accommodate over-dispersion in the counts, the outcome was modeled with a negative binomial distribution.

The fixed component comprised three elements:

(i) **Distributed-lag effect of the main meteorological variable**

$$CB\left(X_i^{(p)}; df_{\text{var}}^{(p)}, df_{\text{lag}}^{(p)}\right),$$

with  $df_{\text{var}}^{(p)} \in [2, 4]$  for the exposure spline and  $df_{\text{lag}}^{(p)} \in [2, 4]$  (Deng et al., 2022; Ge et al., 2025) for the lag spline (maximum lag = 6 months).

(ii) **Sum of distributed-lag effects for the remaining covariates**

$$\sum_{q \in S_p} CB\left(X_i^{(q)}; df_{\text{var}}^{(p)}, df_{\text{lag}}^{(p)}\right),$$

where  $S_p$  denotes the subset of the eight candidate meteorological covariates selected (via exhaustive combination search) to adjust the main variable  $p$ . Each adjuster used the same degrees of freedom as its corresponding main variable.

- (iii) **Seasonal component**  $f_{\text{month}}(M_i)$  is a cyclic spline of calendar month with 12 knots that captures baseline annual seasonality in SFTS incidence that is not fully explained by the meteorological covariates. This term absorbs residual seasonal structure from other seasonal drivers and helps avoid attributing general seasonality to the exposure–lag effects of individual meteorological variables.

Therefore,

$$\eta_i = CB\left(X_i^{(p)}; df_{\text{var}}^{(p)}, df_{\text{lag}}^{(p)}\right) + \sum_{q \in S_p} CB\left(X_i^{(q)}; df_{\text{var}}^{(p)}, df_{\text{lag}}^{(p)}\right) + f_{\text{month}}(M_i). \quad (2.2)$$

The group-level random intercept  $b_{g(i)} \sim \mathcal{N}(0, \sigma_b^2)$  captured cluster-specific baseline differences among the three meteorological groups.

The autoregressive random effect within cities was defined as follows: Let  $c(i)$  be the city identity of observation  $i$ , and  $t(i)$  its calendar month. Define the lag operator  $L_r(i)$  as the observation belonging to the same city  $c(i)$  at month  $t(i) - r$ . The autoregressive component was

$$\omega_i = \sum_{r=1}^3 \gamma_r \left[ \ln\left(Z_{L_r(i)}^*\right) - \eta_{L_r(i)} \right], \quad Z_{L_r(i)}^* = \max(Y_{L_r(i)}, 0.5), \quad (2.3)$$

where  $\gamma_r$  ( $r = 1, 2, 3$ ) were autoregressive coefficients capturing short-term serial correlation within the same city time series; the truncation at 0.5 prevented taking the logarithm of zero. When a required lag  $L_r(i)$  did not exist, the corresponding term was omitted.

To assess the robustness of the model results, sensitivity analyses were performed by varying the degrees of freedom for both the main exposure and lag dimensions in the cross-basis functions for meteorological indicators.

## 2.6. Statistical analysis

Variable selection and clustering analyses were performed using the “Boruta”, “caret”, “party”, “permimp”, and “ClusterR” packages. Qualitative predictive analyses for each algorithm were conducted with the “caret”, “randomForest”, “pROC”, “xgboost”, “dlnm”, “gbm”, “lightgbm”, “shapviz”, and “treeshap” packages. Quantitative risk estimation was carried out using the “gamm4”, “MASS”, “MuMIn”, and “performance” packages. Figures were generated with the “ggplot2” package in R software (version 4.4.3). All statistical tests were two-sided, with 95 % confidence intervals (CIs) and a significance threshold of  $P < 0.05$ .

## 3. Results

### 3.1. Temporal patterns and variable screening outcomes

From January 2010 to December 2024, a total of 1250 SFTS cases were reported across thirteen prefecture-level cities in Liaoning Province. Dalian and Dandong together accounted for nearly 87 % of all reported cases, highlighting the strong



spatial aggregation of SFTS risk. Notably, Dalian experienced a sharp increase in recent years, reaching 151 cases in 2021. Since 2015, the epidemic has shown clear signs of geographic expansion, and over the past five years, this trend has intensified, with all cities reporting cases by 2024 (Fig. 2A). SFTS incidence was highly seasonal, clustering between June and September, and both the intensity and duration of the epidemic period have expanded in recent years (Fig. 2B). City-specific heat maps further highlighted heterogeneity in seasonal patterns and temporal trends across municipalities (Fig. S1).

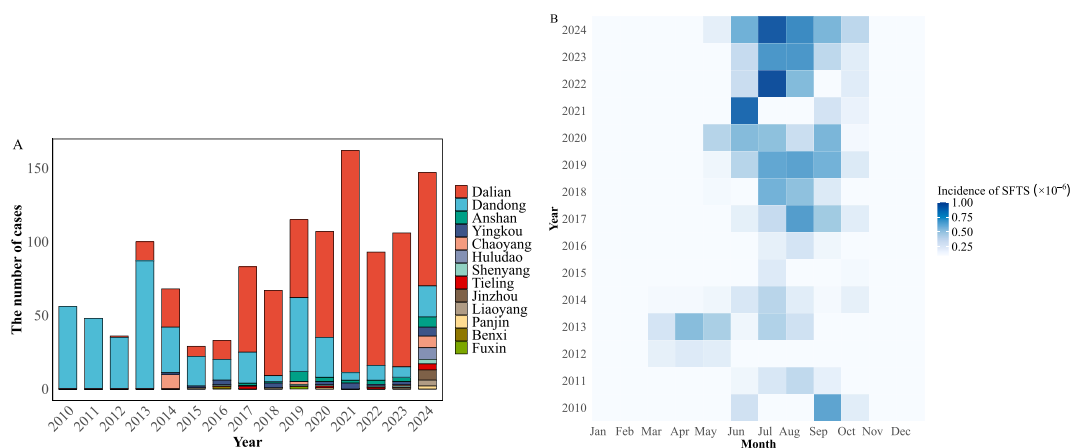
Spearman correlation analysis indicated that SFTS incidence was positively associated with average wind speed (WS), average gust wind speed (GWS), cumulative precipitation (PRCP), normalized difference vegetation index (NDVI), average air temperature (AT), and average land surface temperature (LST), whereas negative associations were observed with average air pressure (AP), cumulative sunshine duration (SD), relative humidity (RH), and the surface–air temperature difference (STD). Among meteorological variables, strong correlations ( $r > 0.8$ ) were found between AT and LST, AT and NDVI, LST and NDVI, LST and STD. Moderate correlations ( $0.7 < r \leq 0.8$ ) were observed for PRCP with AT, LST and NDVI, WS with GWS, STD with NDVI, RH with NDVI, and between various SPEI indices (Table S1; Fig. 3). Based on the correlation analysis, a stepwise VIF screening strategy was adopted to address multicollinearity. Initially, LST (VIF = 729.53) and AT (VIF = 14.36) were sequentially removed due to excessively high VIF values. Subsequently, although the VIF value of NDVI (8.62) was only slightly lower than that of WS (8.94), it was removed first because of its strong correlations with RH and PRCP, which could intensify multicollinearity. Finally, GWS (VIF = 8.85) was removed (Table S2).

To address residual multicollinearity among the screened variables, we further applied Boruta feature selection and conditional permutation importance based on random forest models, with 10-fold cross-validation used to optimize model performance. While Boruta feature selection identified all remaining variables as important (Fig. 4A), the conditional permutation importance analysis revealed a negative contribution for SPEI12 (Fig. 4B), leading to its exclusion. After integrating the importance rankings from both approaches, SPEI3 was also excluded owing to its consistently negligible contribution. Ultimately, nine meteorological variables that were important for SFTS incidence and exhibited low pairwise correlations were retained for subsequent analysis: WS, RH, PRCP, AP, DTR, STD, SD, SPEI1, and SPEI6. Using k-means clustering on the nine meteorological variables, the thirteen prefecture-level cities were classified into three groups: Group 1 (Chaoyang, Huludao), Group 2 (Dandong, Benxi), and Group 3 (Dalian, Yingkou, Jinzhou, Panjin, Shenyang, Tieling, Liaoyang, Anshan, Fuxin) (Fig. S2). According to the PCA loadings (Table S3) and Fig. S2, Group 1 corresponded to a dry–sunny continental climate, Group 2 represented a humid–pluvial orographic climate, and Group 3 was characterized as a maritime–coastal humid climate.

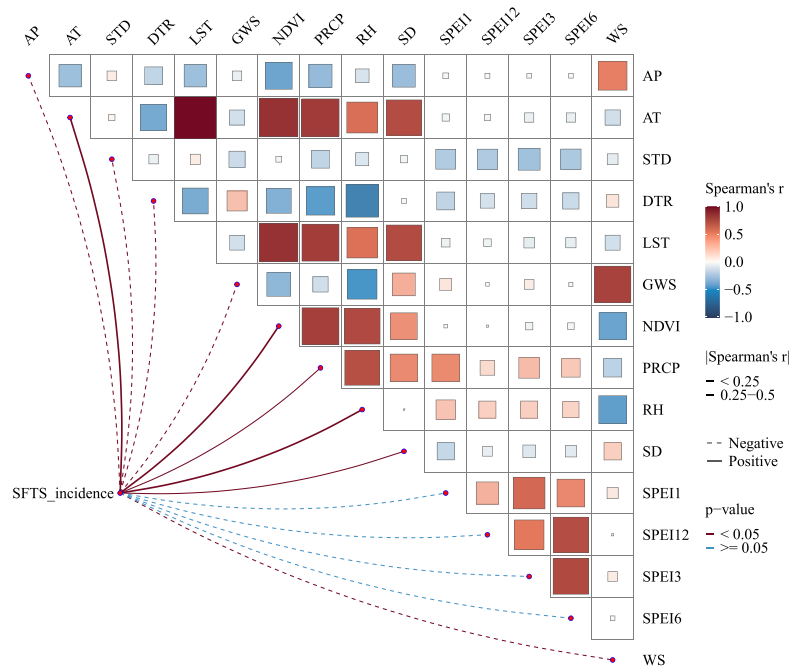
### 3.2. Out-of-time classification performance and SHAP-based insights

Nearly 87 % of all reported cases were concentrated in Dalian and Dandong, whereas the remaining cities contributed only sparse cases. Therefore, we restricted further analyses to these two high-incidence cities. Notably, Dalian and Dandong were assigned to different meteorological groups, highlighting distinct ecological contexts.

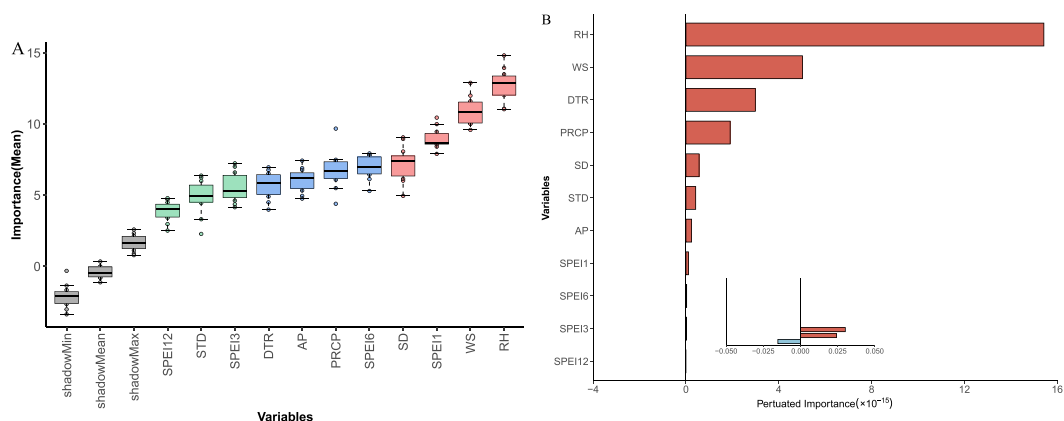
Quantitative prediction models were developed using four independent machine learning algorithms. Robust classification performance was primarily observed in Dalian and Dandong, where case counts were sufficient for reliable modeling. At the same time, results for low-incidence cities were not shown due to limited discriminative information. Overall, random forest (RF) demonstrated the best predictive performance among the four algorithms, achieving the highest AUC (0.935) and recall (0.905), underscoring its strong discriminative ability and high sensitivity in the merged dataset of Dalian and Dandong. This result was important for identifying case months and for minimizing missed detections, especially those



**Fig. 2.** Spatiotemporal distribution of SFTS cases in Liaoning Province, 2010–2024. A: Annual number of reported SFTS cases by city; B: Heat map showing the monthly distribution of cases across years. Incidence was calculated as monthly cases divided by the corresponding city-specific annual population and scaled per  $10^6$  persons.



**Fig. 3.** Spearman correlation coefficients between meteorological factors and monthly incidence of SFTS. Both colors and square size indicate the strength and direction of correlations between meteorological variables (matrix), while line type (solid: positive; dashed: negative) and line color ( $p < 0.05$ : brown,  $p \geq 0.05$ : blue) show the sign and statistical significance of associations between meteorological variables and SFTS incidence. AP: average air pressure; AT: average air temperature; STD: surface–air temperature difference; DTR: diurnal temperature range; LST: average land surface temperature; GWS: average gust wind speed; NDVI: normalized difference vegetation index; PRCP: cumulative precipitation; RH: relative humidity; SD: cumulative sunshine duration; SPEI1, SPEI3, SPEI6, SPEI12: standardized precipitation evapotranspiration index at 1-, 3-, 6-, and 12-month time scales; WS: average wind speed.



**Fig. 4.** Variable importance rankings for meteorological predictors of SFTS incidence. A: Boruta feature selection based on random forest identified all candidate variables as important; B: Conditional permutation importance analysis indicated a negative contribution for SPEI12, supporting its exclusion from subsequent analysis.

occurring outside the peak epidemic period. Extreme gradient boosting (XGBoost) and Light gradient boosting machine (LightGBM) achieved the highest F1 score (0.844) and precision (0.792). Nevertheless, both models yielded areas under the ROC curve that were lower than that of Random Forest, indicating weaker global ranking despite favorable threshold-specific trade-offs. Moreover, the area under the ROC curve for LightGBM was lower than that for XGBoost; therefore, XGBoost yielded the most balanced performance across all metrics. Gradient boosting decision tree (GBDT) underperformed relative to the other three models (Table 1; Fig. 5A–B). Notably, although RF sacrificed some precision, it substantially improved recall to 0.905. The confusion matrices in Fig. 5C further illustrated this advantage: the false negative rate for RF was 4.17 % (2/48). Given the severe outcomes associated with missed SFTS cases and the priority of minimizing false

negatives in early warning systems, the superior recall and low omission rate of RF were particularly valuable. Therefore, RF was selected as the optimal model for subsequent analysis, while XGBoost represented the most balanced alternative.

SHAP analyses for RF models revealed the relative importance of each meteorological variable for SFTS prediction, with the proportional contribution in Dalian and Dandong shown in Fig. 6A. Overall, the RF model yielded very similar importance profiles for Dalian and Dandong, with most meteorological drivers contributing about equally to the mean absolute SHAP values. Modest differences remained: WS weighed more in Dandong, DTR weighed more in Dalian, and STD, SPEI6, and SPEI1 were slightly higher in Dandong. These small deviations implied subtle local sensitivities, while the common pattern indicated stable model behavior across the two cities. The SHAP summary plots in Fig. 6B further illustrated the lag-specific influence and direction of the leading variables, where each  $vi.lj$  term denoted the  $j$ th lagged component of the  $i$ th variable ( $i = 1, 2, 3, 4$  and  $j = 1, 2$ ), the top 36 components were displayed, and the remaining 36 are provided in Fig. S3. For instance, high values of WS\_v2.l2 and WS\_v4.l2 increased risk, while low values were protective; in contrast, WS\_v3.l2 showed the opposite pattern, with lower values associated with higher risk. Similar bidirectional effects were observed for components of AP, PRCP, DTR, and SPEI1. This bidirectional, value-dependent, and heterogeneous pattern suggested complex, nonlinear climate-vector-host interactions operating across multiple temporal scales.

### 3.3. Quantitative meteorology-driven exposure-lag risk estimation with optimal covariates

Using each meteorological variable in turn as the main exposure and exhaustively evaluating all possible combinations of the remaining variables as covariates, nine optimal exposure-covariate configurations were identified (Table 2). In the nine optimal combinations, the most recurrent covariates were average air pressure (AP) (8/9) and cumulative precipitation (PRCP) (7/9), followed by cumulative sunshine duration (SD) and surface-air temperature difference (STD) (6/9), 1-month (SPEI1) and 6-month (SPEI6) standardized precipitation evapotranspiration indices (5/9), then wind speed (WS) and diurnal temperature range (DTR) (4/9); relative humidity (RH) appeared only once. This ordering indicated that SFTS risk was most consistently shaped by atmospheric dynamics (AP, WS), moisture availability (PRCP, SPEI1/6), and surface energy balance (SD, STD), with RH adding little once these processes were controlled. Model fit was modest (marginal  $R^2 \approx 0.17$ – $0.34$ ), with the precipitation-anchored specification yielding the highest marginal  $R^2 \approx 0.34$ , implying that precipitation and its correlates provided the most robust and interpretable contribution to risk.

From Fig. S4 and Fig. S5, the ACF and PACF plots of the nine optimal GAM models without third-order autoregressive terms indicated clear residual dependence, with residual autocorrelation  $r_{AC}(\tau) > 0.15$  at some nonzero lags and partial autocorrelation  $r_{PAC}(\tau) > 0.15$  for some lags. This suggested that the plain GAM specification could not adequately account for temporal autocorrelation in the case data. By contrast, the ACF and PACF plots of the corresponding MGAM models incorporating third-order autoregressive terms showed no evident residual autocorrelation, with  $r_{AC}(\tau) < 0.15$  at for all nonzero lags and  $r_{PAC}(\tau) < 0.15$  for all lags (Fig. S6; Fig. S7). These results confirmed that including third-order autoregressive structure effectively addressed serial dependence and ensured the validity of the MGAM framework for risk estimation.

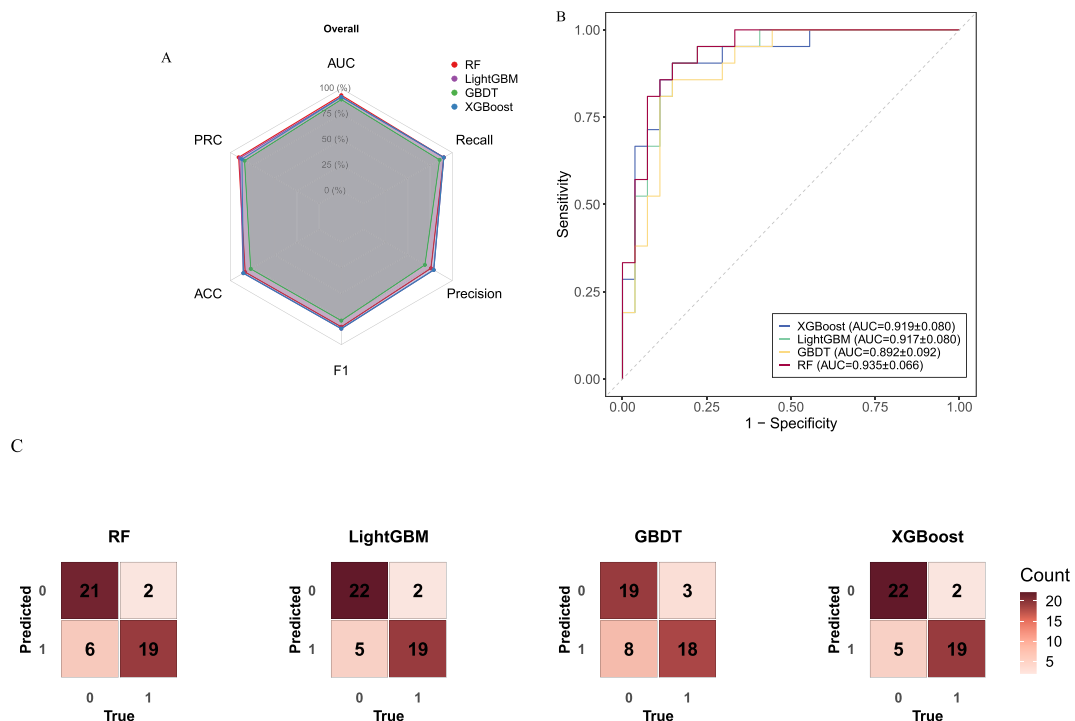
These configurations were subsequently used to examine the heterogeneous effects of each main exposure on SFTS risk (Fig. 7; Fig. 8). Among the nine main exposures, WS, AP, SD, DTR, and SPEI6 exhibited inverted U-shaped relationships with SFTS risk (Fig. 7A–C, F and H). For WS, cumulative risk peaked at 3.54 m/s (RR = 1.185; 95 % CI: 1.008–1.392) across a 0–6 month lag, declining toward both lower and higher speeds, where risks were significantly below 1 outside the range of 3.29–4.16 m/s. AP exhibited a peak relative risk at 990.39 hPa (RR = 1.24; 95 % CI: 1.097–1.402). Risk attenuated as pressure deviated from this point, and the RR was significantly below 1 outside 978.9–997.3 hPa. In the case of PRCP, an intermediate exposure of 46.2 mm corresponded to the maximum risk (RR = 1.32; 95 % CI: 1.158–1.504), followed by a decline in either direction; PRCP below 35.13 mm or above 80.95 mm was associated with risks significantly below 1. DTR reached its risk peak at 10.22 °C (RR = 1.336; 95 % CI: 1.108–1.611), with values beyond 7.48–11.51 °C associated with risks significantly below 1. For SPEI6, maximum risk was observed at 0.32 (RR = 1.313; 95 % CI: 1.031–1.672), with both drier and wetter conditions outside the range from –0.03 to 0.85 associated with significantly lower risk (RR < 1). Distinct from the inverted U-shaped patterns described above, three variables exhibited monotonic associations with SFTS risk (Fig. 7D, E and I). RH and SD exhibited monotonic increases in cumulative risk. Due to data sparsity at the upper extremes, the estimates of these two variables were restricted to values below the 75th percentile. For RH, risk rose steadily, reaching a statistically significant maximum at 68.2 % (RR = 10.132; 95 % CI: 4.910–20.908). SD peaked significantly at 159.92 h (RR = 19.905; 95 % CI: 11.050–35.857). In contrast, STD showed a monotonic decline in risk. To avoid instability from sparse data at the lower end, the estimates of this variable were limited to values above the 5th percentile. The highest and statistically significant risk

**Table 1**

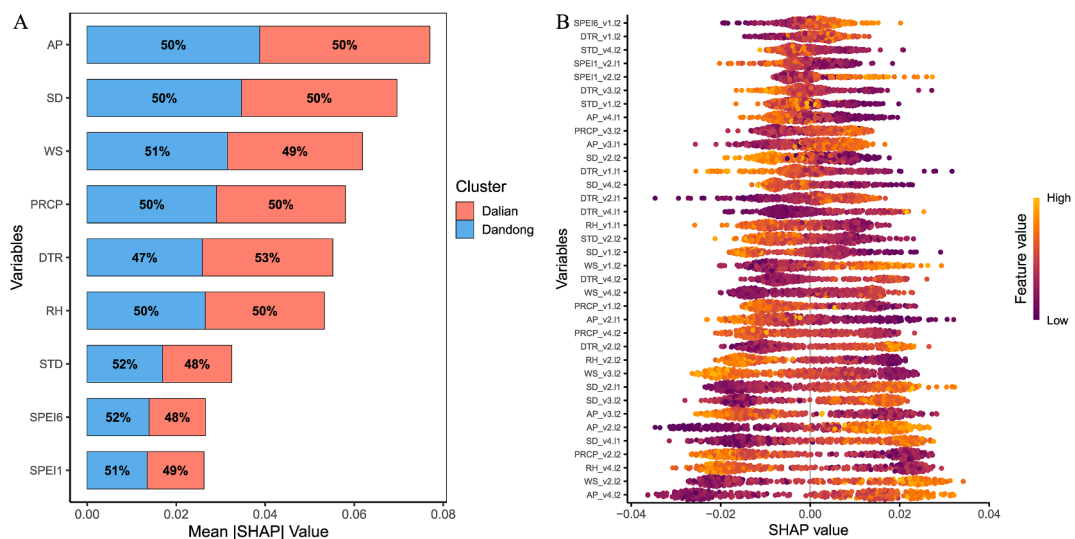
Classification performance of four machine learning algorithms for SFTS occurrence in the pooled monthly dataset from Dalian and Dandong.

| Dalian & Dandong          |                      |           |              |              |              |              |
|---------------------------|----------------------|-----------|--------------|--------------|--------------|--------------|
| Training Obs. (163 zeros) | Test Obs. (27 zeros) | Algorithm | F1 score     | AUC          | Precision    | Recall       |
| 300                       | 48                   | RF        | 0.826        | <b>0.935</b> | 0.76         | <b>0.905</b> |
|                           |                      | XGBoost   | <b>0.844</b> | 0.919        | <b>0.792</b> | 0.905        |
|                           |                      | GBDT      | 0.766        | 0.892        | 0.692        | 0.857        |
|                           |                      | LightGBM  | 0.844        | 0.917        | 0.792        | 0.905        |





**Fig. 5.** Graphical comparison of machine learning algorithm performance for SFTS occurrence in the merged dataset of Dalian and Dandong. A: Radar plot summarizing the main classification metrics for each algorithm; B: ROC curves comparing the discriminative performance of the four models; C: Confusion matrices illustrating classification accuracy and errors for each algorithm.



**Fig. 6.** SHAP-based comparison of variable importance and lagged effects for SFTS risk prediction in Dalian and Dandong using random forest algorithms. A: Average SHAP values illustrated the percentage contribution of each meteorological variable to SFTS prediction in Dalian and Dandong using the RF algorithm; B: SHAP summary plots for RF presented the magnitude and direction of lag-specific contributions for the top 36 variable components.

was observed at  $-1.45^{\circ}\text{C}$  ( $\text{RR} = 5.23$ ; 95 % CI: 3.406–8.034). Surprisingly, SPEI1 exhibited a more complex nonlinear association with SFTS risk, featuring two significant peaks at  $-0.59$  ( $\text{RR} = 1.966$ ; 95 % CI: 1.230–3.144) and  $1.27$  ( $\text{RR} = 4.03$ ; 95 % CI: 1.398–11.616). A significant minimum was observed at  $0.37$  ( $\text{RR} = 0.662$ ; 95 % CI: 0.461–0.951). Owing to sparse data at the upper extreme, the estimates of this variable were restricted to the 95th percentile.

The month-specific effects of meteorological variables varied notably across lag periods from 0 to 6 months. For WS, AP, and PRCP, the strongest associations emerged at the maximum lag of 6 months, with peak relative risks observed at 4.52 m/s

**Table 2**

Optimal meteorological covariate combinations for SFTS risk estimation identified using MGAM, stratified by meteorological main exposure.

| Main exposure | Meteorological covariates             | Exp.df | Lag df | AIC     | BIC     | Marg.R <sup>2</sup> | Cond.R <sup>2</sup> |
|---------------|---------------------------------------|--------|--------|---------|---------|---------------------|---------------------|
| WS            | AP + SD + STD                         | 2      | 3      | 2119.27 | 2297.56 | 0.21                | 0.21                |
| AP            | PRCP + SD + DTR + SPEI1+STD           | 2      | 3      | 2001.93 | 2318.26 | 0.32                | 0.32                |
| PRCP          | WS + AP + SD                          | 3      | 2      | 2417.04 | 2595.34 | 0.32                | 0.34                |
| RH            | WS + AP + SD + PRCP + SPEI1+SPEI6+STD | 3      | 2      | 1968.56 | 2284.89 | 0.29                | 0.29                |
| SD            | AP + PRCP + DTR + SPEI1+STD + SPEI6   | 2      | 3      | 2105.54 | 2387.37 | 0.31                | 0.31                |
| DTR           | AP + SD + RH + PRCP + SPEI1+SPEI6+STD | 2      | 3      | 2092.18 | 2408.52 | 0.29                | 0.29                |
| SPEI1         | WS + AP + SPEI6+PRCP + STD            | 3      | 2      | 2353.00 | 2600.32 | 0.21                | 0.25                |
| SPEI6         | WS + AP + DTR + PRCP                  | 3      | 2      | 2018.86 | 2231.66 | 0.17                | 0.19                |
| STD           | PRCP + AP + SD + DTR + SPEI1+SPEI6    | 2      | 3      | 2105.56 | 2387.38 | 0.31                | 0.31                |

(RR = 1.887; 95 % CI: 1.328–2.682), 1022.72 hPa (RR = 6.756; 95 % CI: 2.316–19.710) and 51.6 mm (RR = 1.061; 95 % CI: 1.000–1.124), respectively. The lowest risk for WS likewise occurred at a 6-month lag, corresponding to 1.72 m/s (RR = 0.059; 95 % CI: 0.024–0.143); whereas for AP and PRCP, the minima were reached at a lag of 0 months, corresponding to 1022.72 hPa (RR = 0.086; 95 % CI: 0.019–0.389) and 0 mm (RR = 0.367; 95 % CI: 0.257–0.532), respectively (Table S4–S6; Fig. 8A–C). In contrast, RH, DTR, SPEI1, and SPEI6 displayed their most pronounced effects at a lag of 0 months, with peak risks corresponding to 89.12 % (RR = 11.208; 95 % CI: 4.686–26.808), 5.04 °C (RR = 4.519; 95 % CI: 1.816–11.246), 2.06 (RR = 3.310; 95 % CI: 1.704–6.432), and 0.52 (RR = 1.198; 95 % CI: 1.080–1.329), respectively. RH also recorded its minimum at the same lag, reaching 21.96 % (RR = 0.429; 95 % CI: 0.154–1.192); while for the other three variables, the lowest risks occurred at 6 months, specifically at 5.04 °C (RR = 0.081; 95 % CI: 0.032–0.207), −3.18 (RR = 0.112; 95 % CI: 0.049–0.254), and −2.75 (RR = 0.357; 95 % CI: 0.176–0.725) (Table S7, S9–S11; Fig. 8D–F–H). For SD and STD, the highest risks were associated with an intermediate lag of 3 months, corresponding to 213.59h (RR = 11.436; 95 % CI: 6.488–20.158) and −3.64 °C (RR = 5.186; 95 % CI: 2.572–10.456), respectively; while both variables reached their minimum risks at a lag of 0 months, at 54.19h (RR = 0.048; 95 % CI: 0.017–0.136) for SD and −3.64 °C (RR = 0.210; 95 % CI: 0.046–0.970) for STD (Tables S8 and S12; Fig. 8E–I). Fig. S8 visualized these patterns as contour plots of the distributed-lag exposure–response relationships (RR) over 0–6 months, with warmer (cooler) colors denoting higher (lower) risk.

The sensitivity analyses (Fig. S9; Fig. S10) confirmed the robustness of the established model, as varying the degrees of freedom in either the main exposure or lag dimensions produced only negligible changes in the results.

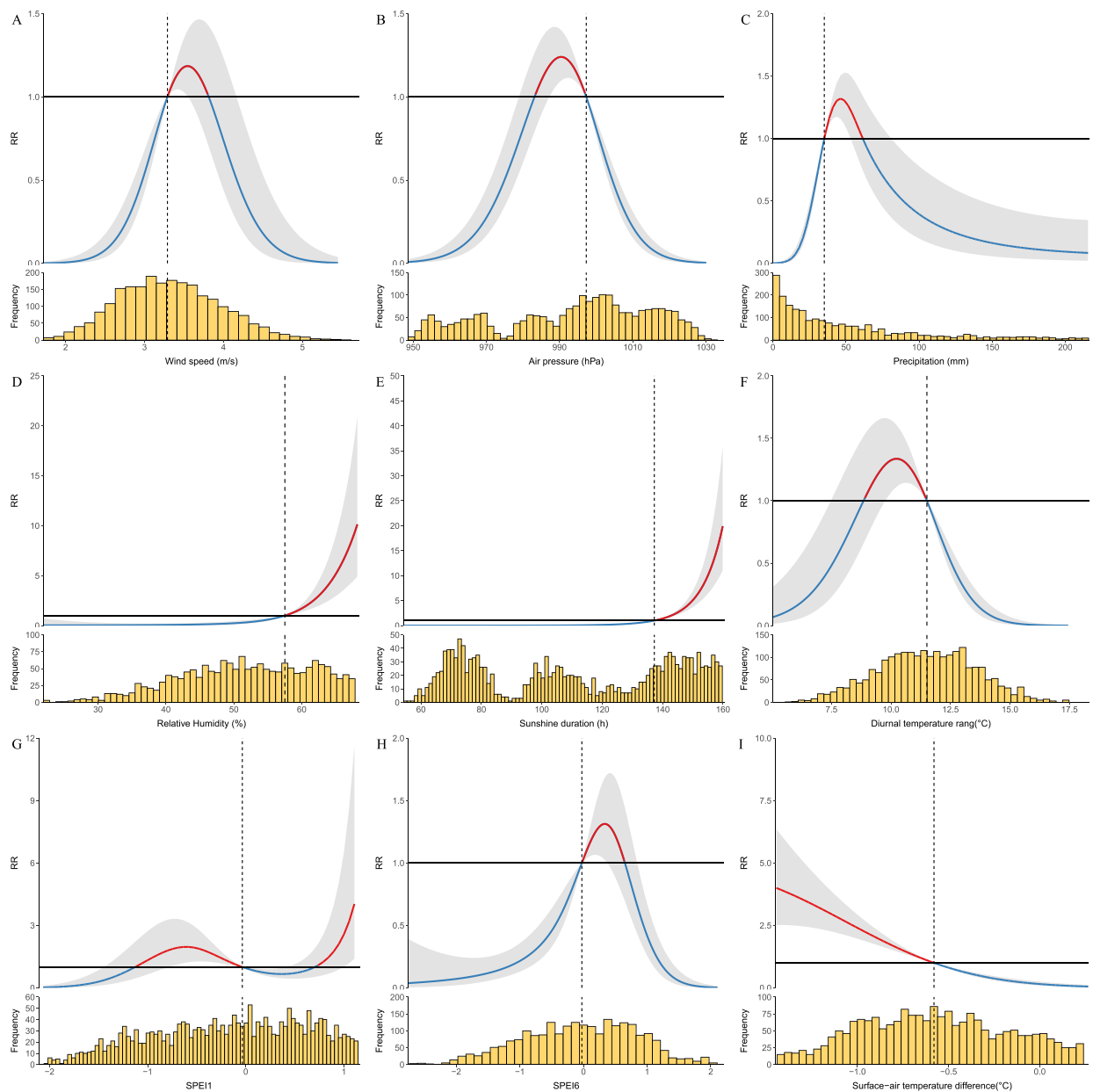
#### 4. Discussion

In this study, SFTS incidence in Liaoning Province showed marked spatiotemporal heterogeneity, with cases concentrated in coastal cities such as Dalian and Dandong, and a pronounced seasonal peak from June to September. To identify key meteorological drivers, variables were initially screened via correlation analysis and variance inflation factor tests, and then refined using Boruta feature selection and conditional permutation importance, yielding nine final predictors. These variables are biologically plausible given their influence on *Haemaphysalis longicornis*, the primary vector in the region: relative humidity (RH) and precipitation (PRCP) reduce desiccation risk, average wind speed (WS) affects dispersal and questing activity, average air pressure (AP) modulates host–tick contact, diurnal temperature range (DTR) and land–air temperature differences (STD) influence microclimatic stability, sunshine duration (SD) shapes habitat suitability, and short-term (SPEI1) and long-term (SPEI6) standardized precipitation evapotranspiration indices reflect vegetation and host availability, together shaping the SFTS transmission cycle.

Using machine-learning binary classifiers, we accurately identified the two SFTS hotspots in Liaoning (Dalian and Dandong) and revealed nonlinear effects of meteorological variables and their lags; for example, different components of WS, AP, DTR and SPEI1 can have opposite effects on SFTS risk (Fig. 6B). Despite broadly similar overall contributions, the RF model captured clear city-specific heterogeneity in several meteorological drivers between Dalian and Dandong. For instance, WS contributed slightly more in Dandong, whereas DTR contributed significantly more in Dalian (Fig. 6A).

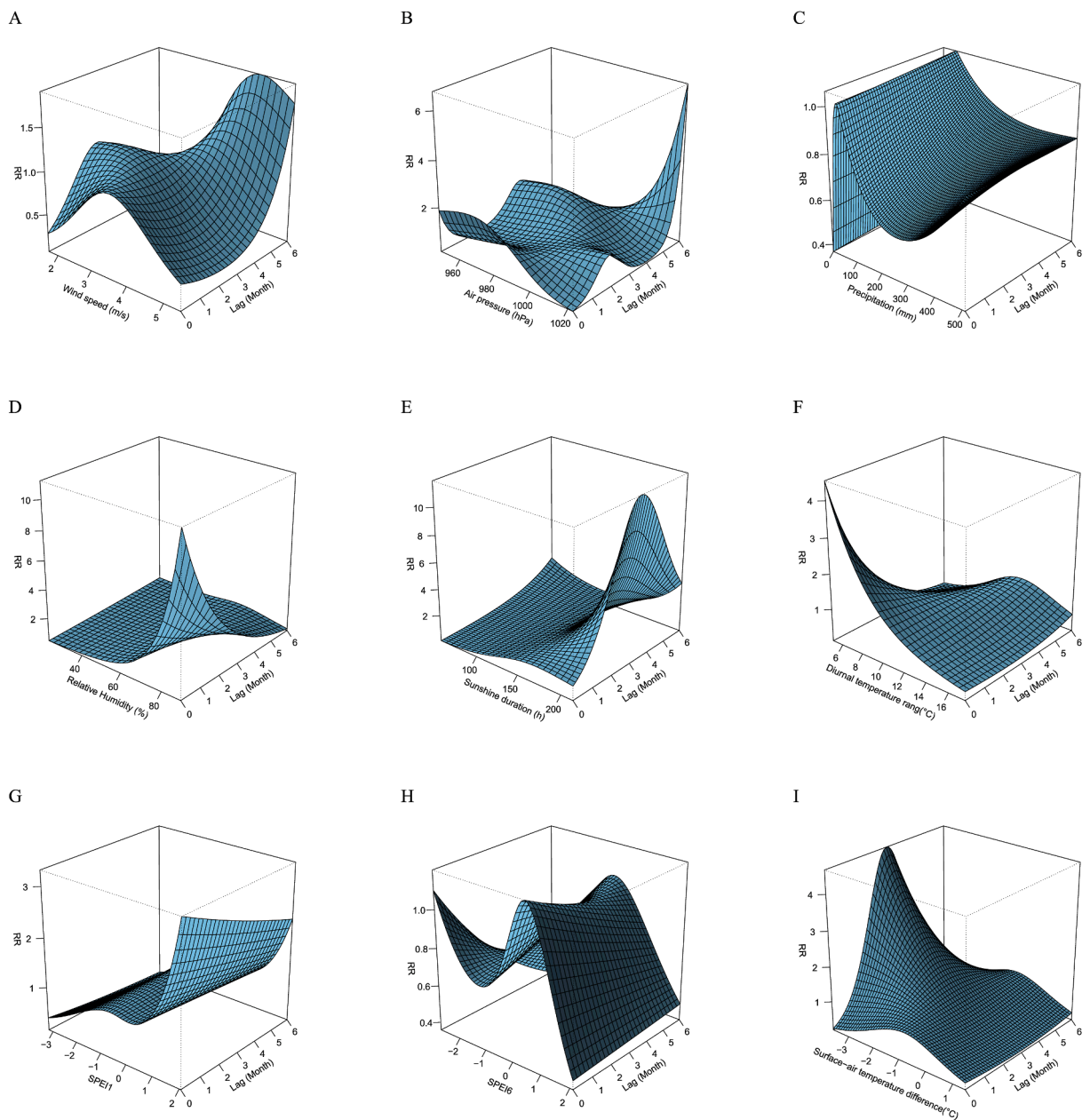
To address the limitations of qualitative analyses in sparsely affected areas, we conducted a quantitative assessment of all cities using mixed generalized additive models (MGAMS) in conjunction with distributed lag nonlinear models (DLNMs), incorporating the k-means clustering results as a random intercept. The resulting optimal combinations successfully identified the risk ranges and significant peaks for each of the main meteorological exposures. In Liaoning Province, the suitable WS range for SFTS occurrence was 3.29–3.81 m/s, with both lower and higher speeds suppressing SFTS risk, consistent with findings from the Jiaodong Peninsula (Wang, Tian, et al., 2024). Previous studies (Garcia, 1962) have shown that sustained wind dispersed carbon dioxide, an essential attractant for ixodid ticks, whereas excessively high wind speeds accelerated the replacement of forest air with drier air from less vegetated areas, thereby increasing tick mortality.

AP, PRCP, DTR, and SPEI6 also exhibited inverted U-shaped associations with SFTS risk. In our study, the optimal AP range for SFTS occurrence was 983.16–997.30 hPa. This may be attributed to moderate decreases in AP, which prolong tick questing activity, lower mortality, and enhance feeding success on natural hosts, thereby increasing the likelihood of human exposure. In contrast, extremely low pressure is often accompanied by adverse weather conditions (such as strong winds and heavy rainfall), which disrupt tick activity rhythms, forcing them to retreat to sheltered habitats and thereby reducing



**Fig. 7.** Cumulative relative risks (RR) of SFTS associated with meteorological factors. A: Cumulative exposure-response effect of average wind speed on the risk of SFTS; B: Cumulative exposure-response effect of average air pressure on the risk of SFTS; C: Cumulative exposure-response effect of precipitation on the risk of SFTS; D: Cumulative exposure-response effect of relative humidity on the risk of SFTS; E: Cumulative exposure-response effect of sunshine duration on the risk of SFTS; F: Cumulative exposure-response effect of diurnal temperature range on the risk of SFTS; G: Cumulative exposure-response effect of SPEI1 on the risk of SFTS; H: Cumulative exposure-response effect of SPEI6 on the risk of SFTS; I: Cumulative exposure-response effect of surface-air temperature difference on the risk of SFTS.

opportunities for host contact (Li et al., 2012). Consistent with our findings (Wu et al., 2020), reported that although SFTS risk generally declined with increasing AP, the peak risk occurred near 1000 hPa. For PRCP, SFTS risk was highest within the range of 35.17–61.46 mm, peaking at approximately 51.6 mm, whereas both drier and wetter conditions were associated with significantly lower risk. Similar inverted U-shaped and V-shaped associations were reported by Jiang et al. (2022) and Sun et al. (2021), with annual PRCP peaking in risk at approximately 600 mm and 1500 mm, respectively. Moderate PRCP within this range may create favorable microclimatic conditions that sustain vegetation moisture and support tick survival, while also promoting host activity; however, prolonged dryness can increase tick desiccation, and excessive PRCP can inundate habitats and physically displace ticks (Jiang et al., 2022).



**Fig. 8.** Three-dimensional lag-response surfaces for the associations between meteorological factors and SFTS risk. A: Lag-response surface of wind speed on the risk of SFTS; B: Lag-response surface of average air pressure on the risk of SFTS; C: Lag-response surface of precipitation on the risk of SFTS; D: Lag-response surface of relative humidity on the risk of SFTS; E: Lag-response surface of sunshine duration on the risk of SFTS; F: Lag-response surface of diurnal temperature range on the risk of SFTS; G: Lag-response surface of SPEI1 on the risk of SFTS; H: Lag-response surface of SPEI6 on the risk of SFTS; I: Lag-response surface of surface-air temperature difference on the risk of SFTS.

Early studies have identified temperature as a key environmental driver of SFTS, typically showing complex nonlinear association with SFTS risk (Du et al., 2014; Jiang et al., 2022; Sun et al., 2021). Temperature affected the life cycle of *Haemaphysalis longicornis*, with both low and high extremes suppressing questing activity and thereby altering the spatial and temporal suitability for tick activity (Wang et al., 2017). However, to our knowledge, the quantitative assessment of DTR on SFTS risk has rarely been documented. Laboratory evidence indicated that moderate temperature fluctuations can maximize tick survival (Randolph, 2004). In our study, the optimal DTR for SFTS occurrence was 8.85–11.51 °C, closely aligning with the findings from Jiangsu Province (8.6–9.7 °C) (Zhang et al., 2019). Such moderate fluctuations likely provided a favorable microclimate for *Haemaphysalis longicornis*, avoiding the stress of persistently uniform or highly variable temperatures. Low DTR may restrict tick questing activity, whereas high DTR can increase desiccation risk. In Liaoning Province, moderate

springtime DTR also coincides with peak host activity, facilitating tick development and virus transmission. A study based on the Jiaodong Peninsula (Wang, Tian, et al., 2024) reported that SFTS risk increased with higher minimum temperatures, which may reduce nighttime cold stress and shift DTR toward the optimal range, indirectly supporting our findings on DTR. Similar associations have been reported for moderate mean or maximum temperatures (Heath, 1979; Namgyal et al., 2020). However, STD showed a monotonic decrease in SFTS risk, suggesting that larger STD, often associated with hotter and drier surface conditions, may reduce vegetation moisture, increase heat and desiccation stress on *Haemaphysalis longicornis*, and suppress their questing activity. Smaller STD indicated more stable and humid microclimates that favored tick survival and host activity, thereby increasing transmission (Rosà et al., 2018).

SPEI6 exhibited a comparable pattern, with the highest risk occurring within the range of  $-0.03$  to  $0.65$ , corresponding to normal to slightly wet conditions. Sustained moderate moisture is likely to provide the most favorable long-term habitat conditions for *Haemaphysalis longicornis* by maintaining vegetation cover, supporting stable host populations, and ensuring the persistence and replenishment of tick populations. In contrast, prolonged dry or excessive wetness can degrade tick habitats or reduce host abundance, thereby restricting opportunities for pathogen transmission (Vail & Smith, 2002). However, SPEI1 displayed a more complex nonlinear association, with two distinct risk peaks separated by a trough. The first peak, occurring when SPEI1 ranged from  $-1.13$  to  $0.04$ , indicated that moderate to mild drought may elevate SFTS risk, possibly attributed to short-term drought reducing vegetation cover, which drives reservoir animals and tick vectors to congregate around limited water sources and even move toward human settlements in search of water (Titcomb et al., 2017). A study in Kenya similarly reported higher tick burdens on calves during the dry season (Chepkwony et al., 2021). The second peak, observed when SPEI1 exceeded  $0.71$ , suggested that mildly to strongly wet conditions can also enhance SFTS risk, possibly through improving vegetation cover and host activity, as well as providing favorable microclimatic conditions for *Haemaphysalis longicornis*. These findings reflected the distinct influences of short-term climatic pulses and long-term ecological background effects, capturing climate–tick–host interaction patterns operating at two different temporal scales.

RH and SD both exhibited positive associations with SFTS risk. When RH exceeded  $57.52\%$ , the risk increased, consistent with the sensitivity of *Haemaphysalis longicornis*, the primary vector of SFTS, to ambient moisture (Tian et al., 2024). Prolonged dry periods can shorten tick survival, whereas higher RH enhanced female oviposition and increased egg hatch rates (Trout Fryxell et al., 2023). However, some studies (Sun et al., 2021) have reported a decline in SFTS risk when relative humidity exceeded  $77\%$ . In our analysis, the scarcity of data in the upper tail of the RH distribution resulted in wide confidence intervals; therefore, we truncated the curve at the 75th percentile, and subsequent risk estimates beyond this range should be interpreted with caution. Similarly, when SD exceeded  $137.26$  h, SFTS risk increased, likely attributed to extended sunlight-enhanced vegetation productivity and host activity, indirectly supporting tick populations. Table 2 further showed the coupling of moisture, energy, and atmospheric dynamics in shaping SFTS risk. AP and PRCP were the most recurrent covariates; lags were short for RH, SD, DTR, STD and SPEI but long for WS, AP, and PRCP, supporting short-term alerts based on RH, SD, DTR, STD, and SPEI1/6, and seasonal outlooks based on WS, AP, and PRCP.

In summary, we implemented a two-stage framework combining qualitative prediction and quantitative risk estimation. In the first stage, we used machine learning classifiers to identify high-incidence areas (Dalian and Dandong) and key meteorological predictors, thereby providing priors for the subsequent stage. In the second stage, mixed generalized additive models (MGAMs) with distributed lag terms and an autoregressive component were fitted, pooling data across cities to address sparse counts. For each meteorological main exposure, we selected a parsimonious, exposure-specific covariate set, chosen sequentially by the Akaike Information Criterion (AIC), the Bayesian Information Criterion (BIC), and statistical significance, to isolate independent exposure lag associations and improve comparability across exposures. Across the two stages, several predictors prioritized by the classifiers, such as AP, WS, SD, and PRCP, also exhibited pronounced nonlinear or lagged associations in the MGAM–DLNM analyses, supporting their relevance for both prediction and inference. In contrast, some variables diverged; for example, RH and SPEI1 ranked lower in the classification stage but showed large cumulative relative risks in the MGAM models, likely because ML importance reflects marginal predictive contribution under correlated inputs, whereas MGAM estimates exposure-specific adjusted effects under the specified lag structure and adjustment set. The results suggest potential for province-level early warning and targeted alerts in Dalian and Dandong, although further evaluation is needed to establish its real-time effectiveness. Integrating these exposure-specific risk windows with biomedical engineering controls, including permethrin-impregnated protective textiles (Vaughn et al., 2014) and host-targeted slow-release acaricide devices (Schulze et al., 2017) such as bait boxes and tick tubes, and scheduling their deployment to coincide with the predicted seasonal peaks can reduce contact between humans and ticks and interrupt SFTSV transmission in identified hotspots. Future work could pair fine-scale risk mapping with low-cost biosurveillance and smart-trap networks (Trout Fryxell et al., 2021). Using microclimate and  $\text{CO}_2$ -lure sensors to automate tick capture and reporting would enable adaptive deployment of repellents, acaricide-treated materials, and reservoir–host interventions aligned with the short- and long-lag signals identified in this study. Furthermore, this two-stage framework could be extended to project future SFTS risk by driving the trained classification and MGAM–DLNM components with projected meteorological series and corresponding future land-use layers, so that future case-month occurrence probabilities and exposure-lag risk windows can be mapped under alternative scenario assumptions; such projections would help anticipate where and when risk may shift across Liaoning.

Several limitations should be acknowledged. Firstly, given that SFTS case data were obtained from clinical surveillance, underreporting is possible, particularly in cities with extremely sparse cases or even no reported cases (e.g., Fushun).



Secondly, to minimize multicollinearity in the integrated modeling framework, average air temperature, an important driver of SFTS transmission, was excluded during variable selection. The retained covariates provide a meaningful and reasonably accurate estimation of SFTS risk, and the two temperature-difference indices further capture its indirect influence. Thirdly, the absence of data on tick density, livestock density, and land cover constrained the incorporation of these ecological determinants into the model. Future studies integrating such data may yield a more comprehensive understanding of the mechanisms underlying SFTS transmission. Finally, due to differences in climatic and geographical conditions, the generalizability of our findings to other regions should be interpreted with caution.

## 5. Conclusion

We proposed a novel two-stage analytical framework that combines qualitative prediction using machine-learning classification with quantitative risk assessment via distributed lag non-linear modeling to elucidate the role of meteorological drivers in shaping SFTS risk. The first stage identified high-incidence areas and key meteorological drivers, capturing regional heterogeneity, while the second stage determined optimal covariate combinations for each main exposure and quantified biologically interpretable exposure–response relationships, even in data-sparse settings. The integration of these complementary approaches enhanced predictive robustness, supported province-level prevention and early warning, and enabled targeted alerts in high-incidence regions.

## CRedit authorship contribution statement

**Ning Yu:** Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Baocheng Deng:** Writing – review & editing, Data curation. **Xue Zhang:** Writing – review & editing, Validation, Supervision, Methodology, Funding acquisition.

## Ethics approval and consent to participate

The use of human data was approved by the Ethics Committee of the First Affiliated Hospital of China Medical University.

## Consent for publication

Not applicable.

## Availability of data and materials

The datasets generated and/or analyzed during the current study are available from the corresponding author, Baocheng Deng, upon reasonable request.

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## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.idm.2026.01.001>.

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